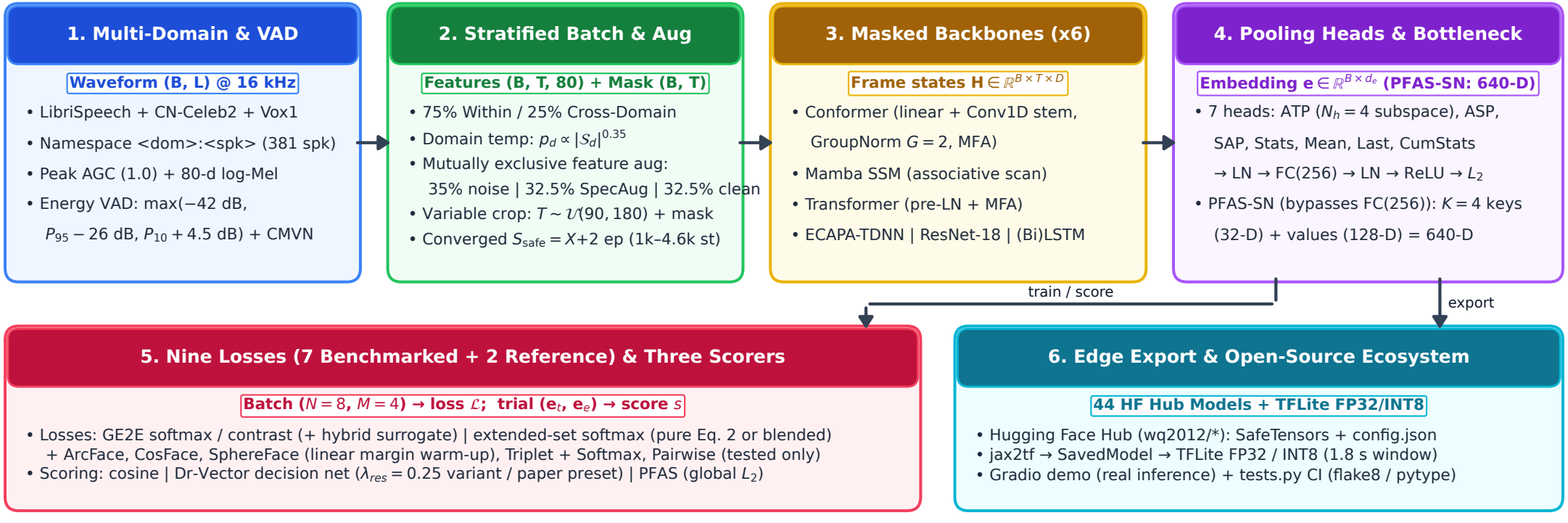

FlaxSpeaker: Multi-Domain Speaker Verification Pipeline (JAX / Flax)
1. Multi-Domain & VAD
Waveform (B, L) @ 16 kHz
• LibriSpeech + CN-Celeb2 + Vox1
• Namespace <dom>:<spk> (381 spk)
• Peak AGC (1.0) + 80-d log-Mel
• Energy VAD: max(−42 dB, $P_{95} - 26$ dB, $P_{10} + 4.5$ dB) + CMVN
2. Stratified Batch & Aug
Features (B, T, 80) + Mask (B, T)
• 75% Within / 25% Cross-Domain
• Domain temp: $p_d \propto |S_d|^{0.35}$
• Mutually exclusive feature aug: 35% noise | 32.5% SpecAug | 32.5% clean
• Variable crop: $T \sim \mathcal{U}(90, 180)$ + mask
• Converged $S_{\text{safe}} = X$+2 ep (1k–4.6k st)
3. Masked Backbones (x6)
Frame states $\mathbf{H} \in \mathbb{R}^{B \times T \times D}$
• Conformer (linear + Conv1D stem, GroupNorm $G = 2$, MFA)
• Mamba SSM (associative scan)
• Transformer (pre-LN + MFA)
• ECAPA-TDNN | ResNet-18 | (Bi)LSTM
4. Pooling Heads & Bottleneck
Embedding $\mathbf{e} \in \mathbb{R}^{B \times d_e}$ (PFAS-SN: 640-D)
• 7 heads: ATP ($N_h = 4$ subspace), ASP, SAP, Stats, Mean, Last, CumStats → LN → FC(256) → LN → ReLU → L_2
• PFAS-SN (bypasses FC(256)): $K = 4$ keys (32-D) + values (128-D) = 640-D
train / score
export
5. Nine Losses (7 Benchmarked + 2 Reference) & Three Scorers
Batch ($N = 8$, $M = 4$) → loss $\mathcal{L}$; trial ($\mathbf{e}_t$, $\mathbf{e}_e$) → score s
• Losses: GE2E softmax / contrast (+ hybrid surrogate) | extended-set softmax (pure Eq. 2 or blended) + ArcFace, CosFace, SphereFace (linear margin warm-up), Triplet + Softmax, Pairwise (tested only)
• Scoring: cosine | Dr-Vector decision net ($\lambda_{res} = 0.25$ variant / paper preset) | PFAS (global L_2)
6. Edge Export & Open-Source Ecosystem
44 HF Hub Models + TFLite FP32/INT8
• Hugging Face Hub (wq2012/*): SafeTensors + config.json
• jax2tf → SavedModel → TFLite FP32 / INT8 (1.8 s window)
• Gradio demo (real inference) + tests.py CI (flake8 / pytype)